

University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 04

Experiment Title: 8-bit ALU Design

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1. Experiment Name : Design and Simulation of an 8-bit ALU Using Structural VHDL and NAND-Based Logic Gates

2. Objective

The objectives of this experiment are:

- To design an 8-bit Arithmetic Logic Unit (ALU) using VHDL.
- To design the required logic gates using NAND gates.
- To construct 8-bit AND, OR, XOR, and NOT operation blocks using previously designed 1-bit gates.
- To implement 8-bit addition using a cascaded 8-bit full adder.
- To implement 8-bit subtraction using the 2's complement method.
- To use a 3-bit selection signal to select the required ALU operation.
- To design a 6-to-1 multiplexer for selecting the final ALU output.
- To implement the ALU using structural VHDL and hierarchical design.
- To simulate and verify the functionality of the designed 8-bit ALU using Xilinx ISE and ISim.

3. Apparatus / Hardware & Software Requirements

Hardware Requirements

- Computer/Laptop
- Keyboard and Mouse
- Minimum 4 GB RAM

Software Requirements

- Xilinx ISE 14.7
- ISim Simulator
- VHDL
- IEEE `STD_LOGIC_1164` library
- Virtual machine environment, if required for running Xilinx ISE

4. Circuit Diagram / Schematic

The 8-bit ALU was designed using a hierarchical structural VHDL approach. The previously designed NAND-based logic gates were reused to construct the required logic operation blocks.

The main operation blocks of the ALU are:

- 8-bit AND
- 8-bit OR
- 8-bit XOR
- 8-bit NOT
- 8-bit ADD
- 8-bit SUB

The outputs of these six operation blocks are connected to a 6-to-1 multiplexer. The multiplexer selects the required operation according to the 3-bit selection signal.

The operation selection is:

SEL[2:0]	**Operation**
000	AND
001	OR
010	XOR
011	NOT
100	ADD
101	SUB

For the arithmetic operations, the previously designed 8-bit full adder was reused.

For addition:

A + B

For subtraction, the 2's complement method was used:

A - B = A + NOT(B) + 1

The SUB control signal is used to invert the bits of B through XOR gates and simultaneously provide the required carry-in of 1.

The RTL schematic generated in Xilinx ISE shows the hierarchical structure of the designed ALU.

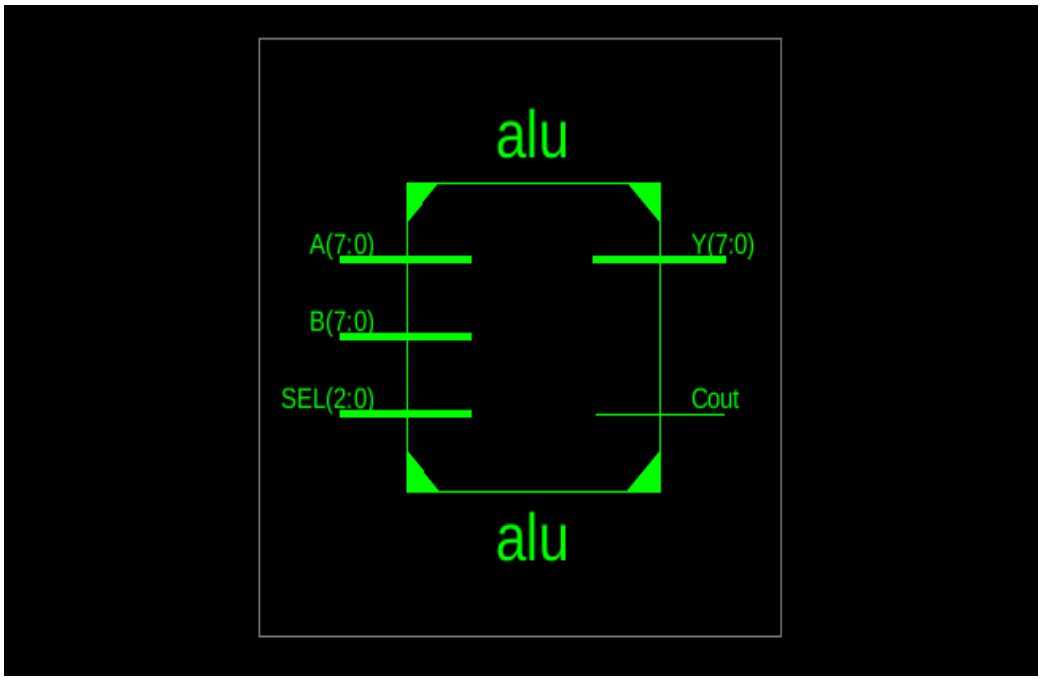


Fig: RTL schematic diagram of the 8-bit ALU

6. Output Analysis / Observations

The designed 8-bit ALU was successfully simulated using the Xilinx ISim simulator.

The following test cases were used:

Operation	A	B	SEL	Expected Y	Cout
AND	10101010	11001100	000	10001000	0
OR	10101010	11001100	001	11101110	0
XOR	10101010	11001100	010	01100110	0
NOT	10101010	00000000	011	01010101	0
ADD	00001010	00000101	100	00001111	0
SUB	00001010	00000101	101	00000101	1
ADD	11111111	00000001	100	00000000	1
SUB	00001111	00001010	101	00000101	1

For the AND operation:

10101010

AND 11001100

10001000

For the OR operation:

10101010

OR 11001100

11101110

For the XOR operation:

10101010

XOR 11001100

01100110

For the NOT operation:

NOT 10101010

01010101

For addition:

00001010 + 00000101

= 00001111

For subtraction:

00001010 - 00000101

= 00000101

The simulation waveform showed that the output `Y[7:0]` changed according to the selected operation. The results obtained from the simulation matched the expected values.

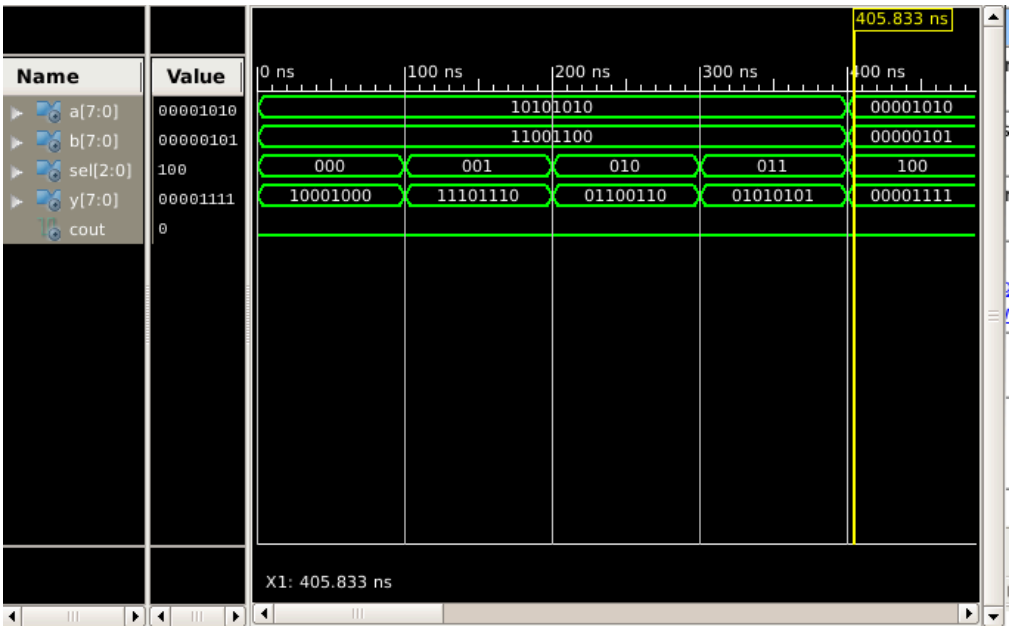


Fig: 8-bit ALU waveform

7. Result

The ALU successfully performed six operations:

- AND
- OR
- XOR
- NOT
- ADD
- SUB

A 3-bit selection signal was used to select the required operation.

The logical operations were implemented using the previously designed NAND-based gate modules. Addition was implemented using the cascaded 8-bit full adder, while subtraction was implemented using the 2's complement technique.

The designed ALU was simulated using Xilinx ISE/ISim, and the observed outputs matched the expected results.

Therefore, the objective of the experiment was successfully achieved.

8. Github Repository Commits

GitHub Repository:
https://github.com/Akifhossain29/ISE-14.7-VLSI

GitHub Commit Link:
https://github.com/Akifhossain29/ISE-14.7-VLSI/commits/main/

9. Conclusion

In this experiment, an 8-bit Arithmetic Logic Unit (ALU) was successfully designed using structural VHDL.

The ALU was capable of performing logical operations such as AND, OR, XOR, and NOT, as well as arithmetic operations such as ADD and SUB.

A 3-bit selection signal was used to select the desired operation. The logical operation blocks were constructed by reusing the previously designed gate modules. Ripple-carry propagation was used in the 8-bit full adder for arithmetic addition, while subtraction was implemented using the 2's complement technique.

Finally, a 6-to-1 multiplexer was used to select the required operation result and produce the final ALU output.

The complete design was simulated using Xilinx ISE/ISim, and the simulation results verified the correct functionality of the ALU.